Causal Concentration–Response Modeling with Continuous Curves and Exposure Error Correction: $PM_{2.5}$ and Mortality in the Medicare Cohort

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Background: Many studies have reported associations of fine particulate matter with aerodynamic diameter ≤2.5 μm ($PM_{2.5}$)with mortality but fewer at low concentrations and even fewer using causal modeling or correcting for exposure error bias. None have corrected for the nonrepresentativeness of monitoring locations.

Objectives: We examined the association of $PM_{2.5}$ with all-cause mortality in the Medicare cohort using a combination of causal modeling, flexible concentration–response modeling, and bias correction for exposure error, while controlling for NO_2 and O_3 as well as standard confounders.

Methods: Using monitors not used to fit our $PM_{2.5}$ model, we fitted 72 regression calibration models stratified by season, region, and elevation in the US. We fitted a B-spline with 4 degrees of freedom to the calibrated $PM_{2.5}$ and fitted separate generalized propensity score models for each spline component using gradient boosting. We also used inverse probability weights to account for the nonrepresentativeness of monitoring locations. Using the generalized propensity scores and the B-splines, we fitted quasi-Poisson models to counts of deaths in each ZIP code-year stratified by race, Medicaid status, and gender. Separate models were fit for participants identifying as black and as white and for ZIP codes with higher and lower poverty rates. We fit a model using the original exposure to estimate the extent of exposure error bias.

Results: The propensity score analysis achieved good balance for all covariates. Controlling for the propensity scores, we found a concentration–response curve with no evidence of a threshold and whose confidence interval did not include the null from 4 $\mu g/m^3$ and upward. There were 223,666,531 person-years of follow-up between the current US Environmental Protection Agency (EPA) standard of 9 $\mu g/m^3$ and the World Health Organization (WHO) guideline of 5 $\mu g/m^3$, and the rate ratio between them was 1.088 [95% confidence interval (CI): 1.064, 1.113]. Using the original exposure, the rate ratio was 1.076 (95% CI: 1.070, 1.083). Hence, effects continue below the EPA standard, and calibrated estimates of effect were 16% higher. Effects were larger from 8 $\mu g/m^3$ among participants identifying as black.

Discussion: The concentration–response curve between air pollution and mortality remains after adjustment for exposure error and using causal models and continues to concentrations below current US EPA and EU standards and even below WHO guidelines. Exposure error in the original exposure resulted in noticeable downward bias at low concentrations. Persons identifying as black are more susceptible. https://doi.org/10.1289/EHP15238

Introduction

The Global Burden of Disease estimates that ambient air pollution is responsible for millions of deaths per year,[1] and the World Health Organization (WHO)[2] has set its new guideline for fine particulate matter with aerodynamic diameter ≤2.5 μm ($PM_{2.5}$) at 5 $\mu g/m^3$. Yet regulators in the US, EU, India, and China have been reluctant to tighten standards, which can be costly and result in industrial opposition. For example, the recent revision of the US Environmental Protection Agency (EPA) annual standard for $PM_{2.5}$ to 9 $\mu g/m^3$, which will take 6 years to be implemented, comes when the current annual average $PM_{2.5}$ in the US is about 8 $\mu g/m^3$, and existing regulations, the continuing retirement of coal electric generation plants and growth of wind and solar generation, and replacement of older vehicles with newer ones with tighter emissions controls and with electric vehicles almost guarantee that these standards will not require reductions in the overwhelming majority of the US. The costs of tighter standards and the observational nature of the epidemiology studies that suggest a tightening of existing standards would be protective of the public's health are major reasons for this reluctance. The observational nature of these studies has resulted in criticism including *a*) the possibility that the associations do not persist at low concentrations,[3] *b*) the dearth of studies using causal modeling methods,[4] *c*) the possibility that measurement error in exposure renders the observed concentration–response relationships unreliable for standard setting, particularly at low concentrations, and *d*) the likelihood of increased exposure error in parts of the country that are less likely to have monitors. In addition, most existing cohort studies have not had adequate sample size to examine disparities in these impacts by race or socioeconomic status, and this is a major public health concern.

Causal modeling methods try to reframe an observational study to mimic a randomized trial.[5] While standard epidemiology seeks to control for confounding by including the confounders in the outcome model, causal methods seek to, in what is called the design stage, make the exposure independent of the confounders, as randomization would do in a trial. If successful, the outcome regression should provide a marginal estimate of the causal effect of exposure under certain assumptions, including no omitted confounders and the stable unit treatment value assumption. If the exposure is independent of the confounders, then we can compare what would have happened had people had high vs. low exposure. Recent papers have used propensity score techniques to examine the concentration–response association of $PM_{2.5}$ and both mortality[6] and cardiovascular disease[7] by dividing exposure into intervals and fitting separate propensity scores by interval. While this approach is indicative of the general shape of the concentration–response curve, it is inherently categorical,

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Supplemental Material is available online (https://doi.org/10.1289/EHP15238).
J. Schwartz declares that he is an Expert Witness for the US Department of Justice in a case involving a violation of the Clean Air Act. No other author has anything to declare.
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Received 28 April 2024; Revised 24 April 2025; Accepted 28 April 2025; Published 10 June 2025.
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and by assigning the same exposure to all observations in a category, induces exposure error. Hence a continuous alternative would be useful.

Additionally, errors in exposure can bias the estimated concentration–response association between $PM_{2.5}$ and mortality.[8] This is of particular importance since the US Environmental Protection Agency[3] has expressed uncertainty about the effect of annual $PM_{2.5}$ levels $<9\,\mu g/m^3$. Consequently, an assessment below that threshold that corrects for this bias becomes crucial for public health considerations.

Measurement error corrections have been applied in only a few cohort studies,[9] and these were limited by not having true models for the measurement error everywhere in the US. Moreover, most existing error correction methods assume that the amount of error does not vary in time or space. However, it is widely recognized that existing space–time models for air pollution estimation perform better in some locations (e.g., low vs. high altitude) and times (e.g., by season, across years) or concentrations than others. To our knowledge, the only paper that used a spatially and temporally varying regression calibration approach for a mortality study was that of Feng et al.,[10] and that paper did not address the shape of the concentration–response relationship. In addition, the measurements used as the gold standard in that paper came from monitors whose distribution is uneven with respect to population density, minority population, and other geographic features. Other regression calibration studies have relied on nonrepresentative monitors or a nonrepresentative population. To address this issue, we introduce an inverse probability of measurement weight into regression calibration.

In this paper, we aim to address those concerns by using a causal modeling method applied to an approach for fitting continuous concentration–response relationships combined with space–time varying regression calibration and correction for the nonrepresentative locations of the monitors, an approach to avoid bias due to exposure error.

Data and Methods

Data

Medicare cohort. We obtained data on all Medicare participants in the United States during 2000–2016 from the Master Beneficiary data of the Center for Medicare and Medicaid Services.[11] Medicare covers over 95% of the population ≥65 years of age in the United States. Participants 65 years of age or older and alive on 1 January of the year following their enrollment in Medicare were entered into the open cohort for survival, and follow-up periods were calendar years. Date of death was obtained from this file. Participants were followed until death or censored by 2016.

For each calendar year, we extracted from the Master Beneficiary file the age, sex, race, ZIP code of residence for that year, and whether they were covered by Medicaid that year for each participant. Age, Medicaid status, and ZIP code were updated annually. Race and sex were self-reported at enrollment. Race was included as a covariate because mortality rates are higher for persons identifying as black than for persons identifying as white,[12] and as they also tend to have higher exposure to $PM_{2.5}$, there is a potential for confounding.[13,14] It was also used as an effect modifier to examine differential susceptibility due to structural racism.[15,16] Race was coded as black, white, and other in this data. This file is publicly available from the Centers for Medicare and Medicaid Services.[11] Using this data, we computed, for each ZIP code, the count of deaths in that ZIP code and year for participants stratified by sex, race, and age.

Covariates. We obtained small area–level social, economic, and housing characteristic variables from the US Census Bureau 2000 and 2010 Census Summary File 3[17] at the ZIP code tabulation–area level (ZCTA). Variables were updated each year between 2000 and 2010 by linearly interpolating between the census years. From 2010 to 2016, we used annual data from the American Community Survey 5-year estimates. The Census variables were percentage of the population living below the poverty level, percentage who identified as black, percentage who identified as Hispanic, percentage owner occupied housing, percentage with less than high school education, population density, median household income, and median home value. In addition, the county-level percentage of people who ever smoked and their mean body mass index scores were obtained from the CDC Behavioral Risk Factor Surveillance Surveys of 2000–2016,[18] which were then assigned to each ZCTA within the county and updated each year. From the Dartmouth Health Atlas,[19] we obtained the percentage of Medicare participants who had a hemoglobin A1c test, a low-density lipoprotein cholesterol (LDL-C) test, a mammogram (biannually), an eye exam, and a visit to a primary care physician for each year. These access-to-care covariates were collected at the hospital service area–level and provided by the 2003–2015 Dartmouth Atlas of Health Data (https://www.dartmouthatlas.org) and assigned to all ZCTAs in that area. Values for the other years were extrapolated from these. Distance to the nearest hospital was calculated from the centroid of ZIP code using hospital locations across the US derived from the 2010 ESRI USA Hospitals ArcGIS database (https://www.arcgis.com/home/item.html?id=f114757725a24d8d9ce203f61eaf8f75). To capture long-term smoking history of Medicare participants in each ZIP code, we used the Medicare data to compute their hospitalization rate for lung cancer by ZIP code for each year, since the log of that rate should be proportional to pack-years (Supplemental Material, "Justification of the use of lung cancer rate as a surrogate for smoking history"). This risks overcontrol because air pollution has been associated with increased risk of lung cancer.[20] Moreover, Di et al.,[21] using a random sample of the Medicare population that had smoking data, showed that that these exposure predictions were not associated with smoking with a p-value of 0.38 for the association of $PM_{2.5}$ with current smoking and 0.52 for smoking history, suggesting that this is not an important confounder. We used annual NO_2 exposure and annual O_3 exposure from previously published machine learning models[22,23] to control for other pollutants. Finally, to account for the potential for omitted confounders that vary over time, we included calendar year as a covariate.

$PM_{2.5}$ Data

$PM_{2.5}$ data was taken from the widely used prediction from Di et al.[24] The approach combines the strengths of satellite-based measurements, weather reanalysis data, land-use regression, and chemical transport modeling (CTM). We incorporated GEOS-Chem and CMAQ CTMs as well as MERRA-2 and CAMS weather simulations. Variables included normalized difference vegetation index (NDVI); tree canopy; impervious surface; surface reflectance; satellite measures of absorbing aerosol index; aerosol optical depth; column NO_2; column O_3; elevation; road density; emission inventory; restaurant density; traffic density; truck density; population density; percentage urban land; light at night; meteorological parameters including mixing height, wind speed, humidity, and vertical velocity; and other spatial covariates as predictors. We used a neural network, a random forest, and gradient boosting, each of which can capture nonlinearity and interactions among predictors in the model. Models were trained on monitored data and parameters selected using 10-fold cross-validation. We ensemble averaged these three machine learners with a nonlinear geographically weighted regression. This allows different weights to be given to different learners in

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different parts of the US and in different concentration ranges. Further details can be found in the published paper, and the data is online at https://sedac.ciesin.columbia.edu/data/set/aqdh-pm2-5-concentrations-contiguous-us-1-km-v1-10-2000-2016. The 1-km grid predictions for each day were averaged for all grid cells whose centroids were within the boundary of each ZIP code, and the daily means were averaged to produce an annual mean exposure. This study was approved by the Harvard School of Public Health Human Subjects Committee as reuse of deidentified administrative data.

Statistical Approach

Conceptualization of the Model

We conceptualize this study as an individual-level study with limited individual covariates and a neighborhood exposure. We take the neighborhood exposures as the true exposures of interest because those are the exposures that are regulated by governments, and the scientific and policy question becomes "how does neighborhood exposure affect mortality rates?" We also take this as our true exposure because using this exposure means that confounders are primarily at the neighborhood level, rather than the individual level. That is, if, e.g., air pollution concentrations were higher in neighborhoods with higher blood pressure, lower-blood-pressure individuals in that neighborhood would still get the "high blood pressure neighborhood" exposure and not an exposure based on their blood pressure. This minimizes the importance of the limited individual covariates for confounding. Our goal then is to reduce or eliminate bias in the estimate of the $PM_{2.5}$ effect on mortality correcting for bias due to error between this true exposure and our modeled exposure and due to the unrepresentativeness of the locations of monitors in the context of a flexibly nonlinear causal model. ZIP code is a bit larger definition of neighborhood than ideal and using a ZIP code average of the regression calibrated exposure reintroduces some exposure error. Since this spatial averaging is a form of spatial smoothing, we believe that most of this error is Berkson and hence will not introduce bias effect sizes into the mortality analysis.

Regression calibration. Exposure variables in all epidemiology studies have some error and hence are estimates of exposure. This error can bias the estimated effect of exposure on health in the study. Regression calibration is an approach to reduce or eliminate that bias in the effect size estimate due to classical exposure error, although it does not eliminate exposure error itself (Supplemental Material, "The role of classical measurement error in the analysis").[25,26] In the regression calibration approach, the estimated exposure used as a predictor in the health study is replaced by the expected values of the true exposure predicted from a regression of true exposure against the estimated exposure and other variables. This expected value of true exposure is obtained from a calibration regression. Regression calibration produces an approximately unbiased estimate of the true relative risk for an exposure under the condition that measurement errors in the reported exposure are nondifferential; that is, they are independent of the disease outcome. Spatial simulation studies of high R^2 but spatially misaligned data indicate that regression calibration did an excellent job of eliminating bias.[27]

Because the distribution of air pollution monitors is not representative of the US population, which can create exposure error in the $PM_{2.5}$ model, we first developed inverse probability of sampling weights for the nonrepresentative distribution of monitoring locations. We used a logistic regression to predict the probability of having at least one $PM_{2.5}$ monitoring site for each ZIP code in each year, given population density, percent of the population who identified as being of black race, percent of the population who identified as Hispanic, percent of the population who identified as Asian, percent of persons ≥65 years of age below the threshold for poverty (all from the US Census), distance between industrial facility and ZIP code centroid and road density (from Open Streetmap, https://www.openstreetmap.org/#map=5/38.01/-95.84), normalized difference vegetation index, and tree canopy (from Landsat, https://landsat.gsfc.nasa.gov/data/). Using this predicted probability, we created stabilized inverse probability weights for each monitoring location. We did this to adjust for the uneven distribution of the monitored locations via inverse probability weights, thus making the weighted set of monitors more representative of all ZIP codes.

Next, we stratified the monitored seasonal averaged $PM_{2.5}$ levels by season, elevation (above or below 75th percentile), and Census division (New England, Middle Atlantic, East North Central, West North Central, South Atlantic, East South Central, West South Central, Mountain, and Pacific regions) resulting in 72 strata. This is to account for exposure error varying by season, elevation, and region. The use of the 72 strata means that the assumption of nondifferential error only needs to hold within strata and is an important difference from previous regression calibration work. We used bootstrap aggregation[28] to fit our models by taking 1,000 bootstrap samples from each stratum and then fitted in each sample a regression calibration model of monitored $PM_{2.5}$ levels against predicted $PM_{2.5}$ at the grid cell level. The predictions were from the cross-validation analysis where predictions were made at the 10% held out monitoring locations for each of the 10 cross-validated samples. We included wind speed, humidity, percentage impervious surface, and calendar year as covariates, and weighted by the inverse probabilities of having at least one monitoring site in the ZIP code that contained the grid cell derived from the earlier logistic regression. The estimated coefficients from the 1,000 models in each stratum were averaged, and these averages for all the strata were used to produce our regression calibrated exposure.

Specifically, for each bootstrap sample in each stratum, we fit a model as follows:

$$(PM_{2.5\ \text{monitoring}}) = \beta_{0jk} + \beta_{1jk}\, PM_{2.5\,\text{modeled}} + \beta_{2jk} Z_{jk} + \varepsilon_{jkl}, \quad (1)$$

where j is stratum, k is bootstrap, and l is observation. Z_{jk} are the covariates mentioned above (wind speed, humidity, impervious surface, and calendar year). We estimated stratum-specific regression calibration parameters $\widehat{\beta_{0j}}$, $\widehat{\beta_{1j}}$, and $\widehat{\beta_{2jk}}$ from the mean of the 1,000 $\widehat{\beta_{0jk}}$, $\widehat{\beta_{1jk}}$, and $\widehat{\beta_{2jk}}$ in that strata.

Calibrated predictions were made as

$$PM_{2.5\,\text{calibrated},k} = \widehat{\beta_{0jk}} + \widehat{\beta_{1jk}}\, PM_{2.5\ \text{modeled}} + \widehat{\beta_{2jk}}\, Z_{jk},$$

and those predictions were then averaged to obtain a calibrated prediction for each season in each stratum. The calendar year average predictions were the average of the seasonal predictions. Calibration was done for seasonal averages rather than annual averages because the calibration between predicted and monitored values varied by season.

In this analysis, we took the monitored value as the gold standard for the average concentration in the 1-km grid cell containing it. We based this on two facts. First, secondary pollutants (e.g., sulfates, secondary organic aerosols, nitrates) made up the majority of $PM_{2.5}$ during the study period, and since these are long range transported regional pollutants, their spatial variability over the scale of 1 km should be small. This is supported by the findings of Eeftens et al.[29] that the mean ratio of $PM_{2.5}$ concentrations between street monitors and urban background monitors was only 1.14 and of Pinto et al.[30] that the annual average $PM_{2.5}$ concentrations within an Metropolitan Statistical Area tended to

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be within a few $\mu g/m^3$ of each other. Both contrasts are on scales much larger than, e.g., 0.5 km. We did the regression calibration at the grid cell level to enable us to apply the estimates to studies with different spatial resolutions of health data ranging from geocoded addresses (for which we would use the grid cell corrected estimates) to studies on, e.g., census tract or ZIP code, where we have aggregated the grid cells to the larger spatial resolutions. We acknowledge that the use of a ZIP code average incurs some additional exposure error in this study. Since assigning everyone a ZIP code average rather than an average for a smaller definition of neighborhood is a form of spatial smoothing this is mostly Berkson error, which does not induce bias in effect size estimates.

Mortality analysis. While polynomial terms can be used to estimate nonlinear concentration–response curves, they suffer from considerable collinearity among the terms. B-splines are a basis of piecewise polynomials and are preferred because the piecewise polynomials are more flexible in estimating the true concentration–response curve, and the support for each spline term is concentrated in different ranges of exposure, reducing collinearity.[31–33] This also increases stability when estimating generalized propensity scores for each term. They also avoid the symmetry inherent in polynomials and are more flexible in letting the data determine the shape of the concentration–response curve then prespecified forms such as the Shape-Constrained Health Impact Function.[34]

We fit a B-spline with 4 degrees of freedom to $PM_{2.5}$ to capture nonlinearity in the concentration–response curve between $PM_{2.5}$ and mortality. The B-spline generates four terms to capture the concentration–response curve, and for each of the four spline terms, we fit a generalized propensity score as a gradient boosting regression with 100 trees and a tree depth of four. Gradient boosting captures nonlinearities in the association of confounders with exposure as well as interactions among the confounders.[28] The covariates in the model were those listed in the covariate section, including other pollutants, as well as age, race, sex, and Medicaid status from the denominator file. Balance was assessed by computing the partial correlation between each B-spline component and each covariate, controlling for the four propensity scores.

We then fit a quasi-Poisson model for the count of deaths vs. the four B-spline terms controlling for the four generalized propensity scores. This approach is equivalent to fitting a Cox regression on individual follow-up, controlling for the same variables.[35] To account for any nonlinearities in the association of the generalized propensity score with outcome we fit B splines with 3 degrees of freedom for each of the four propensity scores. That is, the model fit was as follows:

$$\begin{aligned} log\left(E(Y)\right) = {} & \beta_0 + \beta_1 SPL1 + \beta_2 SPL2 + \beta_3 SPL3 + \beta_4 SPL4 \\ & + bs(gps1) + bs(gps2) + bs(gps3) + bs(gps4) \\ & + offset\Big(\log\left(\text{person-time}\right)\Big). \end{aligned}$$

In the above, SPL1 through SPL4 refer to the four components of the B spline basis for $PM_{2.5}$, and gps1 through gps4 refer to the four generalized propensity scores. The concentration–response curve was computed from the coefficients of those SPL terms, and the significance of the association by a likelihood ratio test compared to a model without the B-spline.

To avoid influence by outliers and to ensure positivity (that is, that every unit of observation could have had any exposure), we restricted analysis to $PM_{2.5}$ concentrations between the 2.5th percentile and 97.5th percentile.

To assess susceptibility by marginalized groups, we created subsets of the data with only the counts of deaths by those identifying as white and only the counts of deaths by those identifying as black. Separate truncations of the $PM_{2.5}$ range was done for subgroups since, e.g., black individuals are more concentrated in cities where concentrations tend to be higher. We fit separate generalized propensity scores in each subset, followed by separate quasi-Poisson models for white individuals and black individuals. We similarly created and analyzed subsets for ZIP codes with low vs. high percentage of the age 65+ population below the poverty level (>13% vs. <13%). Finally, as a further test for the potential for confounding by cigarette smoking, we compared our predictions with and without control for the lung cancer hospitalization rate. Rate ratios for 9 vs. 5 $\mu g/m^3$ were estimated by taking the ratio of the predicted rate at 9 $\mu g/m^3$ to the predicted rate at 5 $\mu g/m^3$. Annual deaths that would have occurred had everyone been exposed to 9 vs. 5 $\mu g/m^3$ were computed as mean annual Medicare population × baseline mortality rate × (RR-1)/RR where RR is the above rate ratio. Likelihood ratio tests compared changes in the log-likelihood to the chi-squared distribution. All analyses were done using the R statistical software version 4.1 (R Development Core Team) and the GBM and GLM packages.

Results

Table 1 shows the distribution of the variables in the study. The study covered over 635 million person-years of follow-up, with 648,185 ZIP code-years. Fifty-seven percent of the person-years were female, and 85% were white. Since causal effects estimates at low concentrations were a major concern, we note that there were 223,666,531 person-years of follow-up between the current US EPA standard of 9 $\mu g/m^3$ and the WHO guideline of 5 $\mu g/m^3$; 21,531,741 person-years of exposure below 5 $\mu g/m^3$; and 1,918,857 person-years of exposure below 3 $\mu g/m^3$. For comparison, the Canadian Census Health and Environment Cohort (CanCHEC) had 36 million person-years of follow-up across all exposures.[36] An average of 47 participants died in each ZIP code in each year, for a bit over 30 million deaths. The mean $PM_{2.5}$ exposure was 9.6 $\mu g/m^3$, and the interquartile range (IQR) was 4.2 $\mu g/m^3$. The mean $PM_{2.5}$ exposure for the population exposed to between 5 and 9 $\mu g/m^3$ was 7.45 $\mu g/m^3$. For NO_2 and O_3, the means and IQRs were 16.3 (IQR 11.4) and 39.5 (IQR 4.1) ppb, respectively.

The mean absolute difference between the calibrated and uncalibrated exposure was 0.15 $\mu g/m^3$, with an IQR of 0.15 $\mu g/m^3$. The slope of regressing the two exposures was 0.996.

The four B-spline bases are illustrated in Figure S1, demonstrating the different ranges of support for the four terms. The regression calibration produced estimated annual exposures to $PM_{2.5}$ that were approximately normal, as shown in Figure 1. As shown previously, the relationship between estimate and monitored exposure appeared linear.[21] The error is additive and not multiplicative (Figure S2). Controlling for the generalized propensity score produced good balance for the covariates as shown in Figure 2 and Table 2 for continuous and categorical covariates, respectively.

The causal concentration–response curve is shown in Figure 3, with the dashed lines indicating 95% pointwise confidence intervals (CIs). The likelihood ratio test for including the B-spline terms for $PM_{2.5}$ yielded a p-value of <0.0001. The concentration–response curve was essentially linear between 5 and 12 $\mu g/m^3$, with declining slopes outside of that range. That range contained 72% of the person-time. There was no evidence of a threshold, down to below concentrations of 5 $\mu g/m^3$, which is the WHO guideline. There were 223,666,531 person-years of follow-up when the mean concentration was between the WHO guideline of 5 $\mu g/m^3$ and the EPA standard of 9 $\mu g/m^3$, with a mean exposure of 7.5 $\mu g/m^3$, and there was clearly a significant effect of $PM_{2.5}$ on

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Table 1. Distribution of variables in the analysis of counts of death and $PM_{2.5}$ concentrations in the US Medicare cohort 2000–2016.

Variable	Overall ($n = 648{,}185$ ZIP code years)
Person-years	
Total	635,665,000
Female	363,481,362
Male	272,183,638
White	542,794,734
Black	53,358,394
Other	39,511,872
Deaths	
Total	30,138,538
$PM_{2.5}$ ($\mu g/m^3$)	
Mean ± SD	9.64 ± 3.17
Median (IQR)	9.56 (4.20)
Count of deaths (per ZIP code year)	
Mean ± SD	46.5 ± 73.1
Median (IQR)	13.0 (53.0)
Person time (per ZIP code year)	
Mean ± SD	981 ± 1,550
Median (IQR)	275 (1,100)
Person time (person-years) with $PM_{2.5}$ between 5 and 9 $\mu g/m^3$	223,666,531
Person time (person-years) with $PM_{2.5}$ <5 $\mu g/m^3$	21,531,741
Person time (person-years) with $PM_{2.5}$ <3 $\mu g/m^3$	1,918,857
NO_2 (ppb)	
Mean ± SD	16.3 ± 9.27
Median (IQR)	14.0 (11.4)
O_3 (ppb)	
Mean ± SD	39.5 ± 3.85
Median (IQR)	39.2 (4.13)
Percent with annual eye exam	
Mean ± SD	67.1 (6.82)
Median (IQR)	67.0 (8.48)
Percent with annual LDL test	
Mean ± SD	78.2 ± 7.48
Median (IQR)	79.2 (8.22)
Percent of women with mammogram within 3 years	
Mean ± SD	63.5 ± 7.48
Median (IQR)	63.9 (9.94)
Lung cancer rate	
Mean ± SD	0.000440 ± 0.00365
Median (IQR)	0.000275 (0.000520)
Proportion age >64 living in poverty	
Mean ± SD	0.109 ± 0.114
Median (IQR)	0.0833 (0.0924)
Population density (persons/ km^2)	
Mean ± SD	1,490 ± 5,610
Median (IQR)	108 (960)
Median house value ($)	
Mean ± SD	158,000 ± 145,000
Median (IQR)	113,000 (106,000)
Proportion of ZIP code black	
Mean ± SD	0.0862 ± 0.168
Median (IQR)	0.0121 (0.0797)
Median household income ($)	
Mean ± SD	48,400 ± 23,000
Median (IQR)	43,600 (22,200)
Proportion owner occupied dwellings	
Mean ± SD	0.723 ± 0.178
Median (IQR)	0.765 (0.184)
Proportion hispanic	
Mean ± SD	0.0873 ± 0.158
Median (IQR)	0.0253 (0.0767)
Proportion highest level of education completed < high school	
Mean ± SD	0.288 ± 0.188
Median (IQR)	0.258 (0.237)
Proportion smokers	
Mean ± SD	0.468 ± 0.0744
Median (IQR)	0.465 (0.0927)
Mean BMI (kg/m^2)	
Mean ± SD	28.2 ± 2.68
Median (IQR)	27.6 (1.51)
Percent annual physicians visit	
Mean ± SD	79.2 ± 6.43
Median (IQR)	80.4 (6.62)
Percent annual HbA1c test	
Mean ± SD	83.0 ± 6.17
Median (IQR)	83.9 (6.37)
Distance to nearest hospital (km)	
Mean ± SD	13.0 ± 12.1
Median (IQR)	10.5 (15.4)

Note: IQR, interquartile range; LDL, low-density lipoprotein; $PM_{2.5}$, fine particulate matter with aerodynamic diameter ≤2.5 μm; SD, standard deviation.

mortality in that range. The rate ratio for being at the EPA standard rather than the WHO standard was 1.088 (95% CI: 1.064, 1.113). Had we used the original exposure, the rate ratio would have been 1.076 (95% CI: 1.070, 1.083). The yearly average population covered by Medicare was 38,636,685 during the study period, with a baseline mortality rate of 0.0484 per person per year. Based on the above rate ratio, had they all been exposed at the level of the US standard, this would have translated into an additional 151,874 deaths per year compared to had they all been exposed at the WHO standard.

Figure 4 shows the concentration–response curves when the model was subset to only include black individuals or only include white individuals. Separate generalized propensity scores were fit in each subset. The concentration–response curve in white individuals clearly continues linearly down to 3 $\mu g/m^3$ or lower, with no sign of a threshold. In contrast, for black individuals, as demonstrated by the confidence intervals, there are few observations below 5 $\mu g/m^3$, and the curve becomes higher than for white individuals at about 8 $\mu g/m^3$ and remains above the effect in white individuals for all higher concentrations.

Figure 5 shows the concentration–response curves fit on subsets defined as high poverty (ZIP codes with percentage of persons below the poverty level at or above the 75th percentile of all ZIP codes) or lower poverty with poverty levels below the 75th percentile. There was little difference in the curves between the two groups, and the confidence interval for the higher poverty group contained both the curve for the lower poverty group and its confidence interval. Figure S3 shows the exposure–response curve with and without controlling for lung cancer rate, and there is little difference, confirming the finding of Di et al.[37] that there is no confounding of this exposure by smoking.

Discussion

We have introduced a novel approach using inverse probability weighting to account for the nonrepresentative locations of EPA $PM_{2.5}$ monitors. We applied this to a stratified regression calibration approach dividing the US into 72 strata with separate calibration models in each stratum (by region, elevation, and season). The use of stratified calibration models ensures that the assumption of nondifferential error only needs to hold within strata, reducing a limitation of earlier regression calibration papers. Moreover, it allows for better correction for exposure error in, e.g., strata with lower average concentrations. Using the regression calibrated data, we have estimated causal concentration–response curves using B splines, in contrast to the use of exposure categories as in Wang et al.,[38] providing a more certain concentration–response curve at lower concentrations. There was no evidence of a threshold at concentrations below 5 $\mu g/m^3$ where we have over

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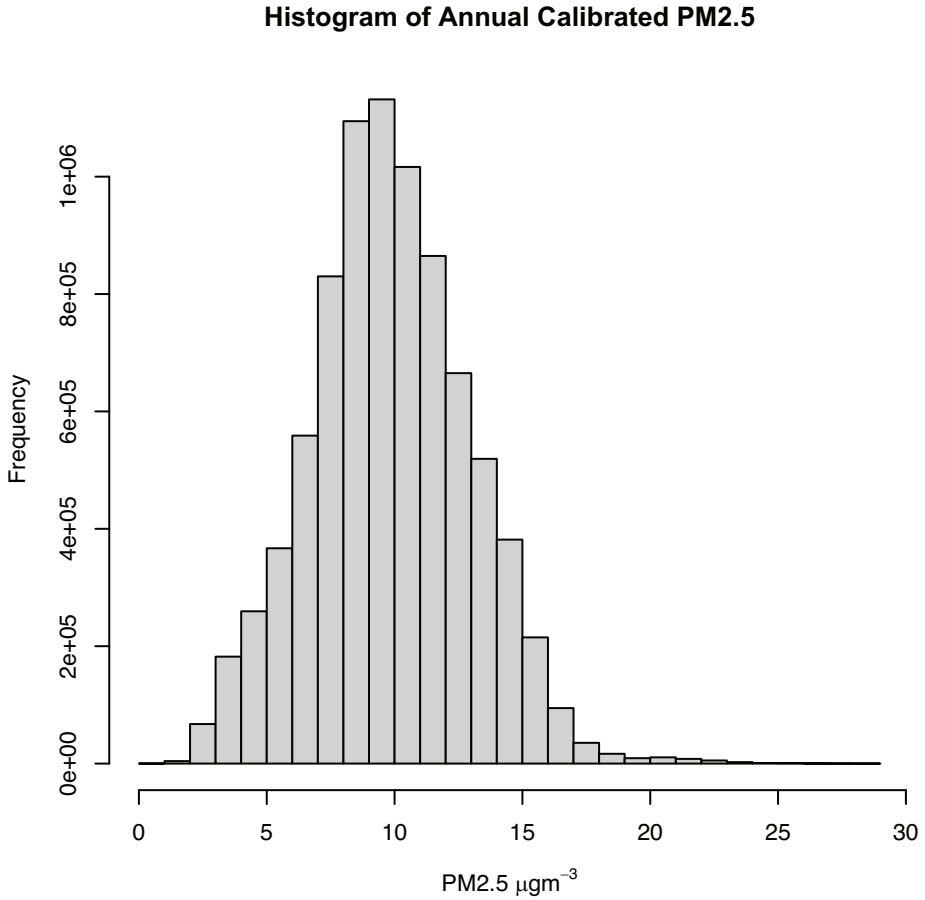


Figure 1. Distribution of annual average regression calibrated $PM_{2.5}$ in the Medicare Cohort, 2000–2016. Note: $PM_{2.5}$, fine particulate matter with aerodynamic diameter ≤2.5 μm.

21 million person-years of follow-up. We demonstrated a substantial difference in mortality rate had the population been exposed to the EPA standard of 9 $\mu g/m^3$ instead of the WHO standard of 5 $\mu g/m^3$ and have shown that the use of the uncalibrated exposure, although with a high R^2 (0.89) on held out monitors,[24] resulted in noticeable downward bias in the effect estimate of $PM_{2.5}$. Our results add to the moderate number of studies showing associations of $PM_{2.5}$ with mortality in three pollutant models, the modest number of studies using causal modeling to control for covariates, the modest number of studies reporting associations below 9 $\mu g/m^3$, and the few studies combining concentration–response modeling with causal modeling. It is, to our knowledge, the only study of $PM_{2.5}$ and mortality combining concentration–response modeling with regression calibration and adjusting for the nonrepresentative distribution of air pollution monitors and the only study using causal modeling and estimating the concentration–response curve with continuous splines rather than using exposure categories. By correcting for the nonrepresentative sampling of locations to site $PM_{2.5}$ monitors in the US, it provides an estimate that should have increased validity in areas of the country where monitoring is less common.

We found that concentrations between the WHO guidelines and the latest US and EU standards are associated with substantial increases in mortality risk. For example, the estimated mortality rate at the US standard was 9% higher than at the WHO guideline. We found that the difference between the Medicare population being exposed at the WHO vs. the EPA standard was an additional 151,874 yearly deaths. For comparison, had the US entirely eliminated deaths from cerebrovascular disease, it would have reduced deaths in 2021 by 165,393 (https://www.cdc.gov/nchs/fastats/stroke.htm). The average concentration of ambient $PM_{2.5}$ in the US during this study was over 9 $\mu g/m^3$, so that is a relevant exposure, and industrial countries such as Finland and Australia as well as many US ZIP codes have already met the WHO guideline, so that is clearly an achievable contrast.

Key aspects of these findings are that the association of $PM_{2.5}$ with mortality continues well below the new US standard and is supported by over 200 million person-years of data below the standard, that correction for measurement error below that standard increases the effect size estimate by 16% rather than making the association null at low concentrations, that achieving the WHO guidelines instead of the US standards would save over 150,000 lives per year, and use of causal modeling in conjunction with measurement error correction also does not render the association null. Hence, the associations of $PM_{2.5}$ with mortality at concentrations below the new EPA standard are robust, not upwardly biased by exposure error, and persist in causal modeling. We also introduced a method for correcting for the nonrepresentative locations of pollution monitors in correcting for measurement error. While a few studies have used regression calibration to correct for exposure error in $PM_{2.5}$,[26,39] they used the regression adjustment populations not representative of the population being studied, and unlike our analysis, assumed a common regression calibration across seasons, locations, and elevation. The study of Feng et al.[10] did use separate regression calibration models by season, location, and elevation but did not correct for the nonrepresentativeness of the EPA monitoring data used for calibration and did not estimate an exposure–response curve.

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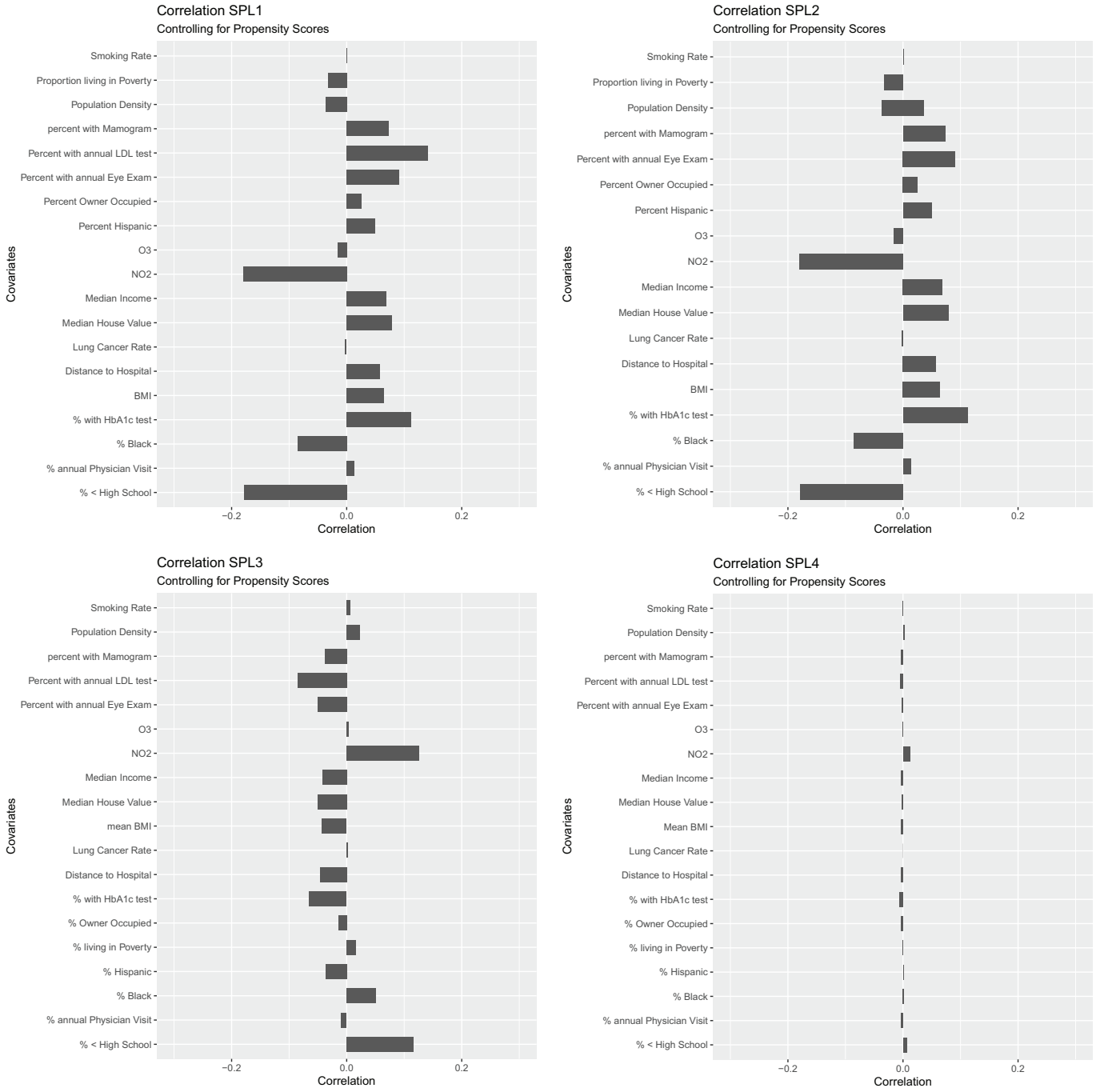


Figure 2. Partial Correlation of B-spline terms for $PM_{2.5}$ with covariates after adjusting for generalized propensity scores in the Medicare cohort, 2000–2016. Numeric results are in Table S1. Note: BMI, body mass index; LDL, low-density lipoprotein; $PM_{2.5}$, fine particulate matter with aerodynamic diameter ≤ 2.5 μm.

Also, the US EPA has expressed concerns about using concentration–response curves in low-exposure ranges when they do not know the mean concentration in that range, and we have provided that. In sum, our study highlights the importance of meeting the WHO guideline as soon as possible and provides evidence that even 5 $\mu g/m^3$ is not a concentration below which there is no effect of $PM_{2.5}$ on mortality.

In analyses stratified by race, we found larger effect size estimates for mortality in black individuals compared to white individuals at concentrations above 8 $\mu g/m^3$. This is consistent with the bulk of the existing literature reporting larger effects among black individuals.[7,40,41] The essentially identical curves by poverty rate differs from previous analyses.

Because we fit separate generalized propensity scores for each of the four B spline components and because each component had most of its support in different ranges of $PM_{2.5}$, our analysis also adjusted for potential confounders differently in different ranges of exposure, which adds robustness to the conclusions.

The finding of an association at such low concentrations has been reported before, particularly in Canadian cohort studies. For example, the Canadian Community Health Survey cohort studied 300,000 people across Canada.[42] The mean annual $PM_{2.5}$ concentration in the participants was only 6.3 $\mu g/m^3$, and 80% of the participants were exposed to concentrations below 8.8 $\mu g/m^3$, yet they reported a similarly strong association with $PM_{2.5}$. Specifically, they reported a 1.26% increase in mortality rate for

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Table 2. Balance for categorical variables: mean residuals (and SD) of each spline term for $PM_{2.5}$ in each category after controlling for propensity scores. US Medicare cohort, 2000–2016.

Race	Mean	SD
SPL1		
Black	−0.008988767	0.1012118
Other	0.005095502	0.1011653
White	0.001479824	0.1016413
SPL2		
Black	0.012813073	0.06883166
Other	−0.005658350	0.08932463
White	−0.002856057	0.08828217
SPL3		
Black	0.005581887	0.06504678
Other	−0.002987355	0.06321809
White	−0.001001227	0.06327491
SPL4		
Black	0.0002314711	0.01842961
Other	0.0002480578	0.01873641
White	−0.0002145366	0.01493031

Note: $PM_{2.5}$, fine particulate matter with aerodynamic diameter ≤2.5 μm; SD, standard deviation; SPL, spline.

an increment of 1 $\mu g/m^3$ in $PM_{2.5}$ vs. our finding of a 2.2% increase. Similarly, a study using the Canadian Census Health and Environment Cohort[36] examined a cohort of 2.5 million Canadians, with 25% having exposures below 6 $\mu g/m^3$ and a median exposure of 8.6 $\mu g/m^3$ and reported a significant association with $PM_{2.5}$. A more recent follow-up with even more data at low concentrations confirmed this.[43] A sequential truncation analysis by Wei et al.[44] using propensity scores reported that the absolute probability of death in the Medicare participants of Massachusetts increased from about 4×10^{-6} to about 6×10^{-6} between exposures <7 $\mu g/m^3$ vs. <9 $\mu g/m^3$. Other studies have reported similar results.[45–48] A common feature of all these studies is of a larger increase in mortality per 1 $\mu g/m^3$ increment in $PM_{2.5}$ than in studies predominantly at higher exposures. Our study supports those results adding causal B spline modeling and correction for exposure error bias and demonstrates that correcting for exposure error results in larger effect size estimates.

The first study in this cohort reported a 13.6% higher mortality rate per 10-$\mu g/m^3$ increment in exposure when restricted to people exposed below 12 $\mu g/m^3$.[21] In comparison, our regression calibrated estimates are an 8.8% higher mortality rate for a 4-$\mu g/m^3$ higher exposure and would have been 7.6% had we used the uncalibrated exposure used by Di et al.[21] Our estimates correcting for exposure error were 16% higher than the uncorrected estimates, but other factors clearly account for part of the difference. One explanation is that the Di et al. paper followed Medicare participants until 2012 whereas this analysis follows them until 2016, incorporating more years with lower exposures. In addition, by using gradient boosting to control for covariates instead of entering them as linear terms, we accounted for potential nonlinear associations between the covariates and $PM_{2.5}$ exposure, as well as interactions among the covariates in our propensity score models. In addition, since each B spline basis function had most of its support in a different range of $PM_{2.5}$ exposure, and separate propensity scores were fit for each basis, we allowed the association between the covariates and $PM_{2.5}$ to differ by $PM_{2.5}$ concentration.

Similarly, some other studies have reported associations of $PM_{2.5}$ with mortality or related morbidities using causal modeling techniques,[45,49–53] including at low concentrations, as well as using methods that control for potential confounding by omitted confounders.[54–56] The conclusion that the observed association in this study is causal relies on three key assumptions: correct specification of the propensity score model, positivity, and no unmeasured confounders. Our use of gradient boosting machines to estimate separate propensity scores for each B spline term with different ranges of support gives some confidence for the former assumption, and the balance plots show very low correlation of those spline terms with all the measured covariates. Our trimming of the exposure interval helps address positivity. While we cannot test whether there are no unmeasured confounders in this study, we can compare our results to other literature that used methods that controlled for most or all unmeasured confounders. For example, a difference in differences analysis in New Jersey reported an effect size of a 3% increase in mortality of persons age 65+ per 2-$\mu g/m^3$ increment in $PM_{2.5}$, similar in size to our results but controlling for all covariates measured or unmeasured that varied across census tracts.[57] The addition of a correction for exposure error bias and B spline concentration–response curve modeling by this paper adds strength to that literature.

Strengths and Limitations of This Study

Our study has several strengths. First, we have addressed the issue of potential measurement error bias in the large number of studies fitting continuous concentration–response curves to $PM_{2.5}$

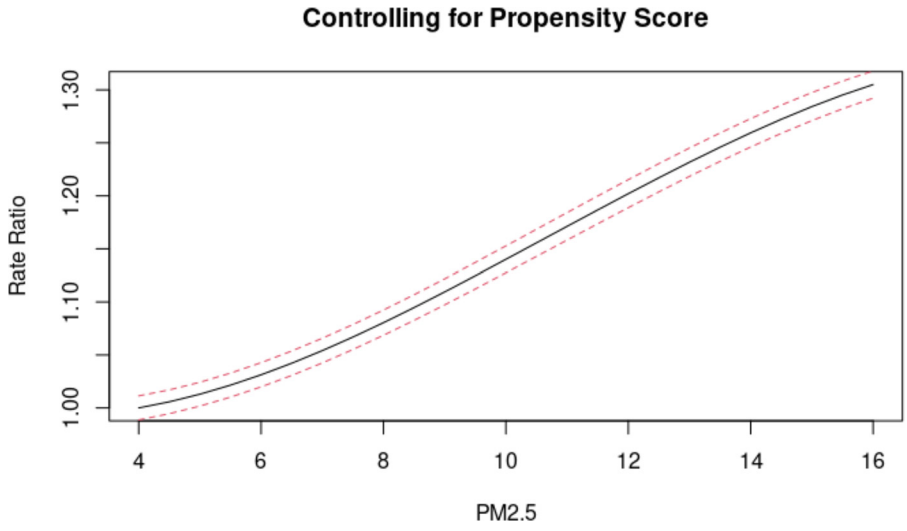


Figure 3. Causal, regression calibrated concentration–response curve between $PM_{2.5}$ and all-cause mortality in the Medicare population: 2000–2016. The dashed lines are pointwise 95% confidence intervals. Note: $PM_{2.5}$, fine particulate matter with aerodynamic diameter ≤2.5 μm.

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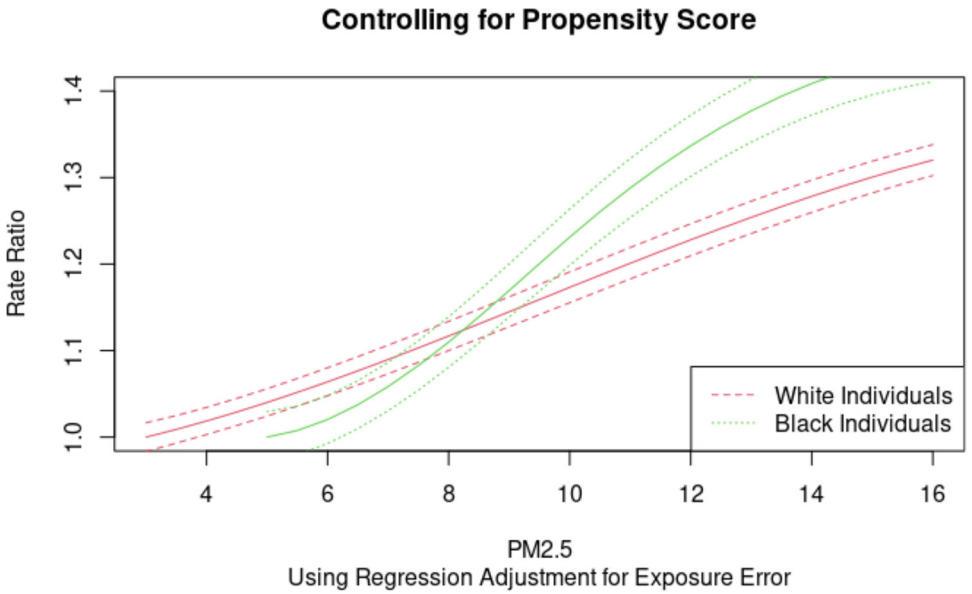


Figure 4. Causal, regression calibrated concentration–response curve between $PM_{2.5}$ and all-cause mortality by race in the Medicare population: 2000–2016. The solid line is for the white population, and the dotted line is for the black population. The dashed lines are pointwise 95% confidence intervals. Note: $PM_{2.5}$, fine particulate matter with aerodynamic diameter ≤2.5 μm.

as a predictor of mortality. Further, our regression calibration approach allows the calibration between true and predicted exposure to vary across regions, seasons, and altitude and controls for important covariates that might predict differences between true and predicted. Moreover, we have done this using a flexible causal modeling approach. This approach, which showed excellent balance for a large number of potential confounders, including other pollutants, allowed us to demonstrate effects of $PM_{2.5}$ on mortality well below the current US and EU standards using a flexible concentration–response curve. It also allowed us to estimate the difference in annual deaths in the US between everyone being exposed at the US EPA standard vs. everyone being exposed at the WHO guideline and show that it is large.

This study also has some limitations. First, the finest spatial resolution we were able to obtain from the Centers for Medicare and Medicaid Services was the ZIP code, with a mean population of 9,696 in 2016. This leaves some exposure error that we were unable to capture in our regression calibration. A recent simulation study suggested that this results in a bias toward the null compared to having exposure at a finer spatial resolution.[8] A second remaining issue is the difference between ambient exposure, which we have used, and personal exposure. In a large cohort such as this, that error is unavoidable. However, we believe that the use of outdoor instead of personal exposure is actually beneficial. As pointed out by Webster and Weisskopf,[58] ambient exposure can be an instrumental variable for personal exposure that avoids confounding by behavioral and other factors. For example, personal exposure is influenced by factors such as more driving or more cooking. However, those behaviors can also influence health through other pathways, such as more

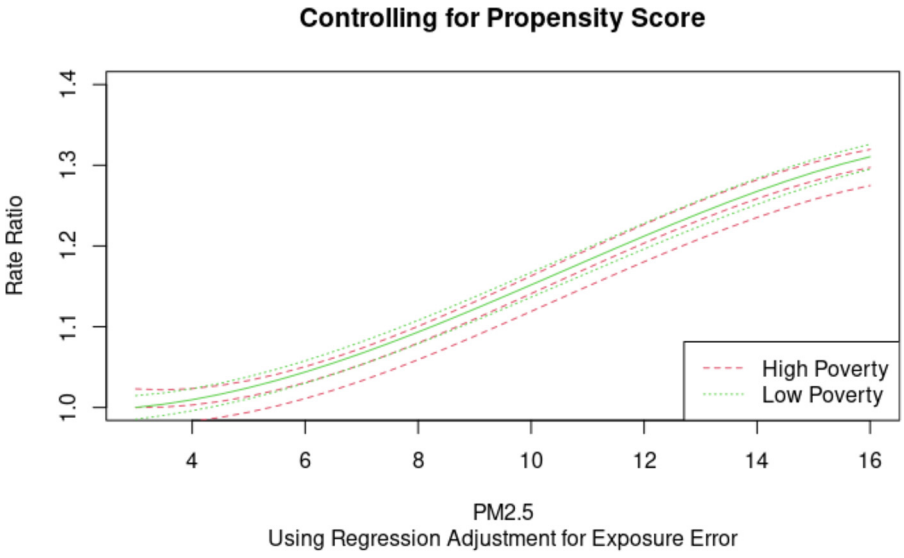


Figure 5. Causal, regression calibrated concentration–response curve between $PM_{2.5}$ and all-cause mortality by poverty rate in the Medicare population: 2000–2016. The solid line is for the 75% of the ZIP codes with the lowest poverty rates, and the dotted line is for the 25% of ZIP codes with the highest poverty rates. The dashed lines are pointwise 95% confidence intervals. Note: $PM_{2.5}$, fine particulate matter with aerodynamic diameter ≤2.5 μm.

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stress from more driving, that would be hard to control for. These behaviors do not confound neighborhood exposure. In addition, the exposure error was somewhat larger at concentrations above 12 $\mu g/m^3$; however, as this paper focused on lower exposures, this is less relevant.

Another limitation is that the Medicare data include few personal risk factors. Here, again, the use of neighborhood exposure is an advantage. Neighborhood air pollution levels are unlikely to be substantially influenced by individual covariates. Ambient pollution concentrations are spatially patterned and, hence, correlated with neighborhood-level not individual-level confounders. That is, while neighborhoods with lower income may also be neighborhoods with higher air pollution level, if people with low incomes were to move to high-income neighborhoods, they would receive an ambient exposure associated with the high-income neighborhood rather than their own individual characteristics. Hence, only neighborhood level covariates are likely confounders, which we have controlled for. Personal exposure is also subject to reverse causation. That is, people who die in a given year may have been ill that year and spent less time in cars and less time cooking than others, and hence, their personal exposure may be lower than people who did not die. Use of ambient exposure avoids this issue.

There are also some limitations based on our analytical choices. Our confidence intervals do not incorporate uncertainty in the estimation of the propensity scores or uncertainty in the regression calibration. We have also assumed that, conditional on the covariates, there is no serial correlation in the errors of the health regression.

Conclusion

$PM_{2.5}$ is predictive of mortality rates for US Medicare participants over 64 years of age with confidence intervals excluding the null from 4 $\mu g/m^3$ and higher. The analysis is causal if the propensity score analysis was adequate and there are no omitted confounders, and the balance analysis suggests that the former is not an issue. Exposure error at concentrations below US EPA standards resulted in a 16% downward bias in effect size estimates that was corrected by regression calibration. The similarity of effect size to studies that controlled for omitted confounders using quasi-experimental methods suggests that the latter is unlikely to be an issue as well. The analysis controlled for exposure error using regression calibration separately by season, region, and elevation, which should reduce or eliminate bias in the effect size estimate due to exposure error. The difference in deaths between meeting the current US EPA standard vs. the WHO guideline is large and indicates that current US and EU standards should be revised toward the WHO guidelines.

Acknowledgments

This research was supported by NIEHS R01ES032418.

The air pollution data ($PM_{2.5}$, NO_2, O_3) used in this study is publicly available from the SEDAC website (https://sedac.ciesin.columbia.edu/data/sets/browse). Medicare data is protected by Data Use Agreements, and interested parties can obtain their own Data Use Agreement from the Centers for Medicare and Medicaid Services to access the data.

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